

122. Given, $\frac{52416}{312} = 168 \Leftrightarrow \frac{52416}{168} = 312$.

Now, $\frac{52.416}{0.0168} = \frac{524160}{168} = \left(\frac{52416}{168} \times 10\right) = (312 \times 10) = 3120$.

123. Given, $168 \times 32 = 5376$ or $5376 \div 168 = 32$.

Now, $\frac{5.376}{16.8} = \frac{53.76}{168} = \left(\frac{5376}{168} \times \frac{1}{100}\right) = \frac{32}{100} = 0.32$.

124. Number of decimal places in the given expression = 8.

Number of decimal places in (a) = 8.

Number of decimal places in (b) = 9.

Number of decimal places in (c) = 7.

Clearly, the expression in (a) is the same as the given expression.

125. For the expressions to be equivalent, the difference between the sum of the decimal places in the numerator and that in the denominator must be equal.

This difference is 1 in the given expression and 1 in (d).

So, (d) is the answer.

126. Given expression = $\frac{(96.54 - 89.63)}{(96.54 + 89.63)} \times \frac{(9.654 + 8.963)}{(96.54 - 89.63)}$
 $= \frac{(96.54 - 89.63)}{(96.54 - 89.63)} \times \frac{(9.654 + 8.963)}{(96.54 + 89.63)}$
 $= \frac{(96.54 - 89.63)}{10(96.54 - 89.63)} \times \frac{(9.654 + 8.963)}{10(96.54 + 89.63)}$
 $= \frac{1}{10} \times \frac{1}{10} = \frac{1}{100} = \frac{1}{10^2} = 10^{-2}$.

127. $(0.11)^3 + (0.22)^3 + \dots + (0.99)^3 = (0.11)^3 (1^3 + 2^3 + \dots + 9^3)$
 $= 0.001331 \times 2025 = 2.695275 \approx 2.695$.

128. $\frac{1}{4} + \frac{1}{4 \times 5} + \frac{1}{4 \times 5 \times 6} = \frac{1}{4} \left(1 + \frac{1}{5} + \frac{1}{30}\right)$
 $= \frac{1}{4} \left(\frac{30 + 6 + 1}{30}\right) = \frac{1}{4} \times \frac{37}{30}$
 $= \frac{37}{120} = 0.3083$.

129. Given expression = $\frac{2 \times 4 \times 8 \times 16 + 4 \times 8 \times 16 + 8 \times 16 + 16 + 1}{2 \times 4 \times 8 \times 16}$
 $= \frac{1024 + 512 + 128 + 16 + 1}{1024}$
 $= \frac{1681}{1024} = 1.6416$.

130. Given expression = $\frac{1}{5 \times 6} + \frac{1}{6 \times 7} + \frac{1}{7 \times 8} + \dots + \frac{1}{24 \times 25}$
 $= \left(\frac{1}{5} - \frac{1}{6}\right) + \left(\frac{1}{6} - \frac{1}{7}\right) + \left(\frac{1}{7} - \frac{1}{8}\right) + \dots + \left(\frac{1}{24} - \frac{1}{25}\right)$
 $= \left(\frac{1}{5} - \frac{1}{25}\right) = \frac{4}{25} = 0.16$.

131. $\frac{1}{5} = 0.2$; $\left(\frac{1}{5}\right)^2 = (0.2)^2 = 0.04$; $\left(\frac{1}{5}\right)^3 = (0.2)^3 = 0.008$;
 $\left(\frac{1}{5}\right)^4 = (0.2)^4 = 0.0016$.

Clearly, for every power which is a multiple of 4, the expression would have 6 as the last digit.

So, $\left(\frac{1}{5}\right)^{2000}$ would have 6 as the last digit.

132. To get 1 as the answer, the expression must have the same number of decimal places in the numerator and denominator.

The number of decimal places in the given expressions are as under:

(a) Numerator $\rightarrow 4$; Denominator $\rightarrow 3$

(b) Numerator $\rightarrow 2$; Denominator $\rightarrow 4$

(c) Numerator $\rightarrow 6$; Denominator $\rightarrow 6$

(d) Numerator $\rightarrow 4$; Denominator $\rightarrow 2$

$\therefore$ The correct answer is (c).

133. Given expression

$$= \left(1 - \frac{1}{4}\right) + \left(\frac{1}{4} - \frac{1}{9}\right) + \left(\frac{1}{9} - \frac{1}{16}\right) + \dots + \left(\frac{1}{81} - \frac{1}{100}\right)$$

$$= 1 - \frac{1}{100} = \frac{99}{100} = 0.99$$

134. $\frac{x}{y} = \frac{0.04}{1.5} = \frac{4}{150} = \frac{2}{75} \Rightarrow \frac{y-x}{y+x} = \frac{1-\frac{x}{y}}{1+\frac{x}{y}} = \frac{1-\frac{2}{75}}{1+\frac{2}{75}} = \frac{73}{77}$.

135. Given expression

$$= 35.7 - \left(3 + \frac{1}{10}\right) - \left(2 + \frac{1}{5}\right) = 35.7 - \left(3 + \frac{3}{10}\right) - \left(2 + \frac{2}{5}\right)$$

$$= 35.7 - \frac{33}{10} - \frac{12}{5} = 35.7 - \left(\frac{33}{10} + \frac{12}{5}\right)$$

$$= 35.7 - \frac{57}{10} = 35.7 - 5.7 = 30$$

136. Given expression = $\frac{(0.3333)}{(0.2222)} \times \frac{(0.1667)(0.8333)}{(0.6667)(0.1250)}$

$$= \frac{3333}{2222} \times \frac{\frac{1}{6} \times \frac{5}{6}}{\frac{2}{3} \times \frac{125}{1000}}$$

$$= \left(\frac{3}{2} \times \frac{1}{6} \times \frac{5}{6} \times \frac{3}{2} \times 8\right) = \frac{5}{2} = 2.50$$

137. Given expression = $\frac{35 \times 1.4}{7} = 5 \times 1.4 = 7$.

138. Given expression = $\frac{16 \times 32}{8} = 16 \times 4 = 64$.

139. Given expression = $\frac{441 \times 16}{21 \times 16 \times 21} = 1$.